Mo | Tu | We | Th | Fr | Sa | Su   Date / /

$y' = x \cdot m + b$ (for data point i: $y_i' = m_1 x_{1i} + m_2 x_{2i} + b_i$)

$J(m, b) = \frac{1}{n} \cdot \Sigma (y_i' - y_i)^2$ if $n=2$ then

$J = \frac{1}{2} [(y_1' - y_1)^2 + (y_2' - y_2)^2]$

chain rule: $\frac{dJ}{dm} = \frac{dJ}{dY'} \times \frac{dy'}{dm}$ | $\frac{dJ}{db} = \frac{dJ}{dY'} \times \frac{dY'}{db}$

$\frac{dJ}{dy'} = \frac{1}{n} \times 2(Y_i' - Y_i) = \frac{2}{n}(y_i' - y_i)$ | let $e_i = (y_i' - y_i)$

$\therefore \boxed{\frac{dJ}{dY'} = \frac{2}{n} \cdot e_i}$

$\frac{dy'}{dm_1} = x_{i1}$, $\frac{dy'}{dm_2} = x_{i2}$ $\therefore \boxed{\frac{dy'}{dm} = x_i}$ (row i of x).

$\frac{dJ}{dm} = \frac{dJ}{dy'} \times \frac{dY'}{dm} = \frac{2}{n} e_i \times x_i = \frac{2}{n} \sum_i e_i x_i$

$\boxed{\therefore \frac{dJ}{dm} = \frac{2}{n} x^T e}$

$y_i' = m_1 x_{1i} + m_2 x_{2i} + b_i$

$\frac{dy_i'}{db_i} = 1$ (since b_i was an added constant)

$$\frac{dJ}{db_i} = \frac{dJ}{dy_i'} \times \frac{dy_i'}{db_i} = \frac{2}{n} e_i \times 1 = \frac{2}{n} \cdot e_i$$

$$\boxed{\therefore \frac{dJ}{db_i} = \frac{2}{n} e}$$

Given $x = [(1,3),(4,10)]$

$m = [-1, 2]$

$b = [1, 1]$

$y = [5, 6]$

$\alpha = 0.01$ (used to minimize overshooting).

$X \cdot m$: row 1: $(1)(-1) + (3)(2) = -1 + 6 = 5$

row 2: $(4)(-1) + (10)(2) = -4 + 20 = 16$

$x \cdot m = [5, 16]$

$y' = x \cdot m + b = [5, 16] + [1, 1] = [6, 17]$.

Let error be e: $e = y' - y$

$= [6 - 5, 17 - 6] = [1, 11]$

$$J = \frac{1}{2}(y' - y)^2 = \frac{1}{2}(1^2 + 11^2) = 61$$

$X^T = [(1,4),(3,10)]$

X^Te: row 1: $(1)(1)+(4)(11) = 1+44 = 45$

row 2: $(3)(1)+(10)(11) = 3+110 = 113.$

$$\boxed{\frac{2}{n}X^Te = \frac{dJ}{dm}} = \frac{dJ}{dy'} \times \frac{dy'}{dm} = \frac{\cancel{2}}{\cancel{2}} \times [45, 113]$$

$$\boxed{\frac{dJ}{dm} = [45, 113]}$$

$$\frac{dJ}{db} = \frac{2}{n}e = \frac{2}{2} \times (1, 11) = [1, 11]$$

$$\boxed{\frac{dJ}{db} = [1, 11]}$$

Gradient descent rule:

new value = old value − (learning rate x gradient)

$$m\text{-new} = m - \alpha \frac{dJ}{dm}$$

$$= [-1, 2] - 0.01\,[45, 113] = [-1-0.45, 2-1.13]$$

$$m\text{-new} = [-1.45, 0.87]$$

$$b\text{-}new = b - \alpha \frac{dJ}{db}$$

$$b\text{-}new = [1, 1] - (0.01)[1, 11] = [1 - 0.01, 1 - 0.11]$$

$$b\text{-}new = [0.99, 0.89]$$